OsteoArthritis and Cartilage (2007) **15,** 98–103
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doi:10.1016/j.joca.2006.06.018

Osteoarthritis and Cartilage

International Cartilage Repair Society

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Brief report
Longitudinal study of magnetic resonance imaging and standard X-rays to assess disease progression in osteoarthritis

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Summary

Objective: To investigate, over 1-year, the relationship between X-ray and magnetic resonance imaging (MRI) findings in patients with knee osteoarthritis (OA).

Methods: Sixty-two osteoarthritic patients (46 women) were followed for 1 year. At baseline and after 1 year, volume and thickness of cartilage of the medial tibia, the lateral tibia and the femur were assessed by MRI. A global score from the multi-feature whole-organ MRI scoring system (WORMS) was calculated for each patient at baseline and after 1 year. This score combined individual scores for articular cartilage, osteophytes, bone marrow abnormality, subchondral cysts and bone attrition in 14 locations. It also incorporated scores for the medial and lateral menisci, anterior and posterior cruciate ligaments, medial and lateral collateral ligaments and synovial distension. Lateral and medial femoro-tibial joint space width (JSW) measurements, performed by digital image analysis, were assessed from fixed-flexion, postero-anterior knee radiographs.

Results: One-year changes in medial femoro-tibial JSW reach 6.7 (20.5) % and changes in medial cartilage volume and thickness reach 0.4 (16.7) % and 2.1 (11.3) %, respectively. Medial femoro-tibial joint space narrowing (JSN) after 1 year, assessed by radiography, was significantly correlated with a loss of medial tibial cartilage volume ($r = 0.25$, $P = 0.046$) and medial tibial cartilage thickness ($r = 0.28$, $P = 0.025$), over the same period. We found also a significant correlation between the progression of the WORMS and radiographic medial JSN over 1 year ($r = -0.35$, $P = 0.006$). All these results remained statistically significant after adjusting for age, sex and body mass index.

Conclusion: This study shows a moderate but significant association between changes in JSW and changes in cartilage volume or thickness in knee joint of osteoarthritic patients.
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Key words: Magnetic resonance imaging, X-ray, Osteoarthritis, Structural progression.

Introduction

Radiography is currently the most widely used method to assess damage in osteoarthritis (OA)[1]. This technique also allows the measurement of joint space width (JSW). New regulatory requirements for the development of disease-modifying drugs in OA still consider the measurement of joint space narrowing (JSN) (a change in JSW over years) on plain X-rays as the appropriate primary endpoint for demonstration of efficacy[2,3]. However, radiography neither allows detecting early structural damage nor constitutes an efficient way of monitoring the progression of OA in daily practice[4,5]. Moreover, radiography poorly assesses other features of OA. Actually, OA is an episodical inflammatory disorder of synovial joints characterized by the focal deterioration and abrasion of articular cartilage, with sclerosis and cyst formation in the underlying bone, as well as formation of osteophytes at the joint surface. Thickening of the joint capsule and chronic synovitis are also commonly present features. Magnetic resonance imaging (MRI), with its superior soft tissue contrast, is the best technique available for assessment of normal articular cartilage and cartilage lesions. Joint imaging has the potential to provide morphologic information, such as the presence of fissuring, partial or full thickness cartilage defects and signal changes within residual cartilage[2,4,6]. Moreover, MRI, with its ability to discriminate articular tissues, holds the greatest potential as a tool for whole-organ imaging of the joint. For this purpose, an MRI global knee joint score has recently been presented[7].

We hypothesize that changes in X-ray are correlated with changes in MRI, in imaging of knee joint of osteoarthritic patients. Indeed, JSW assessed by X-ray has been previously suggested as a surrogate for cartilage loss. The objective of our study was to investigate the relationship between the evolution of severity of MRI (cartilage volume and thickness as well as global joint score) and radiographic (JSW) features.

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Received 25 October 2005; revision accepted 13 June 2006.

Patients and methods

PATIENTS

The cohort consisted in 62 patients with knee OA (Kellgren and Lawrence grading scale of 2 or 3) followed for 1 year. All patients were classified grade 2 or grade 3 at baseline. Main exclusion criteria were history or active presence of other rheumatic diseases that could be responsible of secondary OA, medical or surgical treatment on the knee within the last 6 months, obesity (weight $>$ 250 lbs) and any MRI contraindications.

METHODS

X-ray acquisition

Each knee was radiographed individually using the non-fluoroscopic fixed-flexion technique (Fig. 1)[8,9]. Reproducible positioning of knees on serial examinations was facilitated with a specially designed positioning frame (SynaFlexerä°, Synarc, Inc.) equipped with a projection phantom. The frame positioned the feet reproducibly in 10° external rotation and fixed the degree of flexion of both the tibia and the femur by requiring patients to touch the anterior wall of the frame with their toes, knees and thighs. The postero-anterior X-ray beam was centered on the popliteal fossa at the joint line and angulated 10° caudad. This design ensured that the same region of articular cartilage was projected in the joint space on serial radiographs. Image magnification was corrected using the frame's projection phantom, which was composed of two columns of metallic beads separated by known distances. Training of the X-ray technicians was performed prior to beginning the study by a central radiology service (Synarc, Inc.). All radiographs were visually checked for image quality and proper anatomical positioning by trained personnel at the central radiology service. Deterioration in articular structure was assessed by measuring the change in the narrowest JSW of the medial and lateral tibio-femoral compartments. These measurements were determined semi-automatically by the central radiology service using specialized, validated software (SynaVuä, Synarc, Inc.)[8,10]. Radiographs were digitally scanned and enhanced to improve cortical delineation and computer-assisted JSW measurement. These digitized images were also visually checked for quality and re-scanned, if necessary. Measurement points generated by the automated system were verified and corrected if necessary to ensure that the articular surfaces were properly delineated and that JSW measurements were restricted to the weight bearing regions of the joint rather than, for example, spaces between marginal osteophytes. Serial images of individual patients were measured alongside each other, but were blinded to the order in which they were acquired. The image magnification factor was calculated as the ratio of the projected to known distances between the metallic beads embedded in the phantom on the positioning frame. This magnification factor was then used to adjust the JSW calculations.

MRI acquisitions

MRI of the study knee of each patient was acquired with a 1.5 T whole-body scanner using a commercial circumferential knee coil. Imaging sequences included, axial T1-weighted spin-echo (SE: 700/11[repetition time (TR) ms/time of echo (TE) ms], 20 cm field of view [FOV], 5 mm/1 mm [slice thickness/interslice gap], 256 $\times$ 192 matrix, frequency encoding [FE] anterior–posterior, one excitation), coronal T1-weighted SE (600/11, 16 cm FOV, 4 mm/0.5 mm, 256 $\times$ 192, FE superior–inferior, two excitations averaged), sagittal T1-weighted SE (600/11, 16 cm FOV, 4 mm/0.5 mm, 256 $\times$ 92, FE anterior–posterior, two excitations averaged), sagittal T2-weighted fast spin-echo (FSE: 2500/90, echo train length (ETL) of 8, 14 cm FOV, 4 mm/0 mm, 256 $\times$ 192, FE superior–inferior, two excitations averaged) with fat suppression (frequently-selective presaturation), and sagittal fat-suppressed T1-weighted three dimensional (3D) spoiled gradient echo (FS-3DSPGR: 58/6, 40° flip angle, 14 cm FOV, 256 $\times$ 128 matrix, 60 contiguous 2-mm slices covering all articular cartilage plates in the knee, FE, superior–inferior, one excitation, frequency-selective fat saturation, superior–inferior saturation bands to minimize pulsation artifacts). The total time required for MRI, including patient setup, was 60 min.

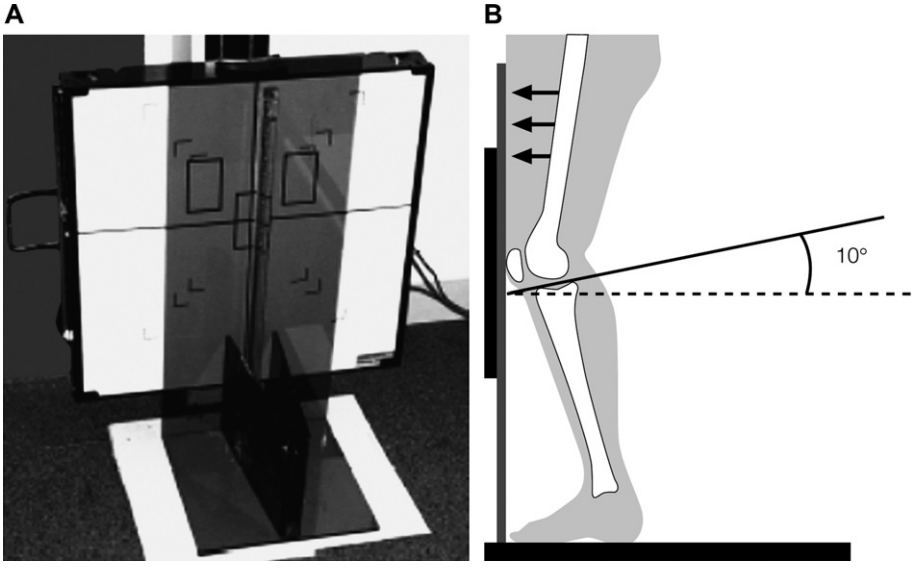


Fig. 1. Fixed-flexion technique.

Volume and thickness assessment

Cartilage volume and thickness measurements of the femur, the medial tibia and the lateral tibia were assessed using a geometrically constrained region growing algorithm, and voxel counting. The segmentations were performed by a trained physician with prior experience in cartilage volume assessment using the iMorphics software (iMorphics, Manchester, UK). Cartilage thickness was evaluated based on a 3D euclidean transformation using the medial axis of the 3D model to define normal as implemented in the iMorphics software.

WORMS

All images were transferred to a Sun Workstation and analyzed using MRVision software (MRVision, Inc., Menlo Park, CA). Images were scored with respect to 14 independent articular features: cartilage signal and morphology, subarticular bone marrow abnormality, subarticular cysts, subarticular bone attrition, marginal osteophytes, medial and lateral meniscal integrity, anterior and posterior cruciate ligament integrity, medial and lateral collateral ligament integrity, synovitis, loose bodies and periarticular cysts/bursae. Two musculoskeletal radiologists performed readings independently. The final WORMS scores (Table I) were tabulated as (1) independent values for each feature in each of the three compartments of the knee [patello-femoral joint (PFJ), medial femoro-tibial joint (MFTJ), and lateral femoro-tibial joint (LFTJ)], (2) cumulative surface feature (cartilage, marrow abnormality, subarticular cysts, bone attrition, and osteophytes) scores for each compartment, (3) cumulative scores for each feature throughout the knee, and (4) a total combined score for the entire knee.

Statistical analysis

Quantitative variables at baseline were expressed as mean $\pm$ SD and qualitative as frequencies. Relationships between MRI (cartilage volume, cartilage thickness and WORMS) and radiological (JSW) severity of knee OA were assessed by the calculation of Pearson correlation coefficients, by a multiple regression analysis and by a stepwise regression analysis. The area under ROC curve measuring the value of medial tibial cartilage volume change over 1 year for the prediction of the radiographic damage (JSN $>$0.3 mm) progression was also calculated. We have taken 0.3 mm as a cut-off for JSN (on X-ray) because of a recent report showing an increased risk of subsequent knee surgery for patients having experienced a JSN of 0.3 mm after 3 years[3]. We have also assessed the discriminative power of a cut-off of volume change maximizing sensitivity and specificity to predict changes in radiographic severity. For this purpose, we have performed analysis of positive and negative predictive values. The same analysis was performed for WORMS progression. The results were considered significant at the 5% level ($P < 0.05$). Statistical calculations were carried out with the Statistica and SPSS software.

Table I
Maximum scores attainable with WORMS

Variable	MFTJ	LFTJ	PFJ	S region	Total
Cartilage	30	30	24		84
Marrow abnormality	15	15	12	3	45
Bone cysts	15	15	12	3	45
Bone attrition	15	15	12		42
Osteophytes	35	35	28		98
Menisci	6	6			12
Ligaments					3
Synovitis					3
Total					332

MFTJ is the medial femoro-tibial joint; LFTJ is the lateral femoro-tibial joint; PFJ is the patello-femoral joint; S region is the non-articulating portion of the tibial plateau beneath the tibial spines.

Results

Baseline characteristics of the population are summarized in Table II. At baseline, lateral femoro-tibial JSW was significantly correlated with lateral tibial cartilage volume ($r = 0.48$, $P = 0.0001$) and thickness ($r = 0.58$, $P = 0.0001$). Medial femoro-tibial JSW was significantly associated with medial tibial cartilage volume ($r = 0.38$, $P = 0.002$), medial tibial cartilage thickness ($r = 0.55$, $P = 0.0001$) and cartilage thickness in femur ($r = 0.26$, $P = 0.039$). All these results remain statistically significant after adjusting for age, sex and body mass index.

One-year changes in medial femoro-tibial JSW reach 6.7 (20.5) % ($P = 0.03$) and changes in medial cartilage volume and thickness reach 0.4 (16.7) % ($P = 0.76$) and 2.1 (11.3) % ($P = 0.23$), respectively. Lateral femoro-tibial JSN, over 1 year, was not correlated with loss of cartilage or thickness at any site. However, medial femoro-tibial JSN after 1 year was significantly correlated with a loss of medial tibial cartilage volume ($r = 0.25$, $P = 0.046$) and medial tibial cartilage thickness ($r = 0.28$, $P = 0.025$), over the same period. After adjusting for age, sex and body mass index, medial femoro-tibial JSN after 1 year was correlated with medial tibial cartilage volume loss ($P = 0.014$) [Table III(A)]. The area under ROC curve measuring the value of medial tibial cartilage volume change over 1 year for the prediction of the radiographic damage (JSN $>$0.3 mm) progression is reported in Fig. 2. Tibial cartilage volume loss over 1 year is a highly significant predictor of the risk to present a structural progression on the X-ray within a period of 1 year ($P = 0.023$). A loss of 5% in cartilage volume is the cut-off value maximizing sensitivity and specificity. The discriminative power of a cut-off of 5% decrease in medial tibial cartilage volume to predict a loss of 0.3 mm in medial JSW is reported in Table IV. When considering loss of at least 5% in cartilage volume, the negative predictive value is 88%, e.g., a patient without such a significant loss of medial tibial cartilage volume at the MRI has less than 12% risk to present a significant loss of JSW (over 0.3 mm) on the X-ray.

Table II
Baseline characteristics of the patients

Variables	Mean (standard deviation)
Age (year)	64.9 (10.3)
Body mass index (kg/m^2)	28.1 (4.7)
Lateral JSW (mm)	5.6 (1.4)
Medial JSW (mm)	4.1 (1.2)
Lateral tibial volume	1742 (528)
Medial tibial volume	1301 (337)
Femur volume	6316 (1231)
Lateral tibial thickness	1.83 (0.35)
Medial tibial thickness	1.52 (0.22)
Femur thickness	1.17 (0.15)
WORMS	43.8 (34.9)

Table III
Multivariate analysis results for the prediction of 1-year structural changes

Variable	Parameter estimate	Standard error	*P*-value
(A) In medial tibial cartilage volume			
Gender	−0.02	0.12	0.85
BMI	−0.13	0.13	0.33
Age	−0.09	0.13	0.47
One-year changes in medial JSW	0.32	0.12	0.014
Medial tibial cartilage volume at baseline	−0.21	0.12	0.09
(B) In WORMS			
Gender	0.14	0.12	0.25
BMI	0.24	0.13	0.08
Age	0.16	0.14	0.28
One-year changes in medial JSW	−0.37	0.12	0.004
Baseline WORMS	−0.09	0.12	0.44

We found a significant correlation between the progression of the WORMS and radiographic medial JSN over 1 year ($r = -0.35$, $P = 0.006$). This result remained statistically significant after adjusting for age, sex and body mass index ($P = 0.004$) [Table III(B)]. No significant correlations were observed between the progression of the WORMS and lateral JSN. The area under ROC curve measuring the value of WORMS change over 1 year for the prediction of JSN >0.3 mm is 0.73 (95% CI 0.56−0.89; $P = 0.013$). An increase of at least one unit in WORMS has been associated with a worsening (0.3 mm and above) in radiological JSW with corresponding sensitivity and specificity of 57% and 82%, respectively (Table IV).

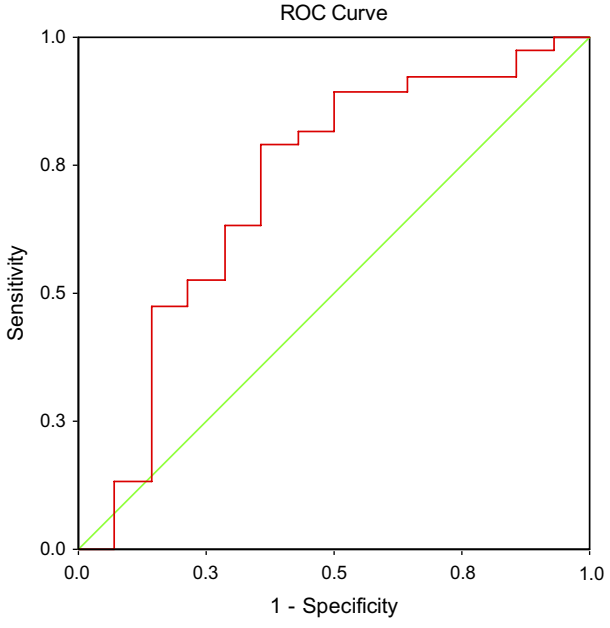


Fig. 2. Area under ROC curve measuring the value of medial tibial cartilage volume change over 1 year for the prediction of JSN >0.3 mm.

Discussion

In this study, we have shown a significant association of MRI and X-rays findings, at baseline, for the assessment of knee OA. Few studies have compared MRI (i.e., cartilage volume or thickness) and X-rays' severity (i.e., JSW, osteophytes, …)[11–16]. The presence of radiographically diagnosed osteophytes at the tibio-femoral joint has been previously reported to be associated with a high prevalence (sensitivity of 77% and specificity of 83%) of a cartilage defect detected by MRI in the same joint[12]. In 50 patients with knee OA, the Kellgren and Lawrence score, assessed by radiography, correlated significantly with the grade of cartilage lesions, assessed by MRI[13]. A semi-quantitative (range from 0 to 3) assessment of patello-femoral JSN on a X-ray was found to be negatively correlated with the volume of the cartilage of the patella[14]. However, no significant associations were found between patella cartilage volume and osteophytes measured on radiographs[14]. Another study shows a strong negative association between medial and lateral tibial cartilage volumes and increasing grade of JSN[15]. The same authors reported that early (grade 1 on the radiograph) medial OA was associated with substantial reductions in cartilage volume, assessed by MRI[16].

All these studies assessing, at baseline, the relationship between X-ray and MRI structural severity of OA are generally based on radiographic structural change scores only. Measurement of JSW is more sensitive to change and more reliable than scores, to evaluate OA severity on a radiograph[2]. JSW can be measured manually (with a caliper, a graduated plastic ruler or a graduated magnifying lens) or with a computer after digitalization of the radiographs. In our study, JSW was measured with a validated computer-assisted method. With this method, we have been able to demonstrate, for the first time, a significant positive association between tibial cartilage volume and thickness, assessed by MRI, and femoro-tibial JSW, assessed by standard radiography.

In this study, we have also shown a significant association between the change over time in MRI and standard X-rays assessment of OA. These results are important because measurement of changes in JSW is currently the gold standard in evaluation of structure modifying drugs in OA[2,3]. To the best of our knowledge, very few studies have dealt with this topic. Cicuttini *et al.* reported, over a period of ±2 years, that JSN, measured on lateral patello-femoral radiographs significantly correlates with loss of patella cartilage, assessed by MRI[17]. Another team does not show any statistical correlation between loss of cartilage volume and radiographic changes, in 32 patients with knee OA[18]. In the latter study, however, the absence of significant association could be explained by the relatively moderate number of patients and by the radiographic protocol. In fact, the use of the standardized

Table IV
Discriminative power of various cut-off of changes in medial tibial cartilage volume and thickness to predict a loss of 0.3 mm in medial JSW

Variables	Sensitivity (%)	Specificity (%)	Positive predictive value (%)	Negative predictive value (%)
One-year cartilage volume loss ≥5%	67	76	48	88
At least one unit changes in WORMS	57	82	50	86

semiflexed knee radiographs, used in the study, has been showed to have poor reproducibility in longitudinal studies[19]. It should be also pointed that the cartilage volume or thickness changes over 1 year do not reach statistical significance in this study. It could be due to the matrix size that we have used in this study. To increase matrix size, amongst a number of factors, would provide a more accurate measurement of cartilage loss and could then modify our results. This size was chosen to minimize time, and thereby patient discomfort.

Our results are one of the first to provide evidence of an association between MRI and X-rays' changes in the femoro-tibial joint. Based on the ROC curve, the negative predictive values of our suggested threshold (i.e., the probability of no significant joint space loss on X-ray in a patient without cartilage loss over the cut-off value) are acceptable (80−86%), but the positive predictive values (i.e., the probability of significant joint space loss on X-ray in a patient with cartilage loss over the cut-off value) remain marginal (39−43%). However, the limited number of patients included in this study, as well as the small duration of this study could limit the validity of these results. Larger and longer studies are required to establish the ROC curve for cartilage volume and JSN. Anyway, our results reemphasize the fact that JSW assessed on standard X-ray, is only an indirect assessment of cartilage loss and that more data on MRI should be provided before accepting it as primary endpoint in clinical phase III trial on disease-modifying OA drugs. However, it should be noted that the use of MRI as a potential endpoint technique is reemphasized by the fact that rate of cartilage loss over 2 years, assessed by MRI, predicts subsequent total knee arthroplasty[20]. The same group has also shown that there was a trend toward a significant association between change in medial tibiofemoral cartilage volume and joint replacement at 4 years, but not change in medial tibio-femoral JSW[21].

In the present study, we have also assessed joint damage by a whole-organ MRI instrument. Already 50 years ago, a whole-organ tool (assessed on standard X-ray) was presented by Kellgren and Lawrence. This was interesting and important because OA involves cartilage but also bone and synovium. Since that time, other whole-organ X-ray tools were validated. However, standard X-ray does not detect changes in soft tissue contrast. MRI, with its superior soft tissue contrast, holds the greatest potential as a tool for whole-organ imaging of the joint. So, it could be possible to extend the idea of the whole joint X-ray scoring to MRI. In this study, we used a validated whole-organ MRI score (WORMS)[6]. Using this technique, we found a significant correlation between the progression of the WORMS and radiographic medial JSN over 1 year. However, no significant relationships were found with lateral JSN, maybe because of the highest prevalence of medial femoro-tibial OA in our study population. These results are interesting but other studies (e.g., correlation between WORMS and clinical symptoms) are needed to assess the clinical relevance of these findings and help us better understand the results of this trial. Actually, it should be noted that this was a 1-year study, with a limited number of patients, and that larger and longer studies are required to conclusively establish the association between cartilage volume changes and WORMS changes (i.e., ROC curve). Moreover, WORMS score assesses features that are not highly sensitive to change (e.g., bone attrition, ligament integrity, …), especially in a 1-year study. A number of larger and longer studies, such as the NIH Osteoarthritis Initiative are currently underway. At least, it should be pointed out that, albeit they have been initiated, no studies have showed an increased risk of knee surgery, in patients with an increased MRI knee score (WORMS or any other).

We have also shown an absence of association between WORMS changes and lateral JSW changes. This reemphasizes the importance of medial, not lateral, JSW in the progression of knee OA.

In conclusion, we have shown a significant association between MRI and standard X-rays' assessment of progression of OA. In particular, we have shown that changes in femoro-tibial JSN after 1 year were associated with loss in tibial cartilage volume or thickness. However, these relationships are only moderate and of limited value for the individual patient. It should be pointed that our study is of short duration (as the majority of studies dealing with these topics) and that X-ray/MRI relationships may well be stronger in longer duration studies. Also, our results emphasized that X-ray does not exactly reflect the 3D structure of the knee joint. At least, more studies are needed to validate MRI as a surrogate for X-rays' JSN in phase III clinical studies of disease-modifying OA drugs.

References

1. Felson DT, Zhang Y, Hannan MT, Naimark A, Weissman BN, Aliabadi P, *et al*. The incidence and natural history of knee osteoarthritis in the elderly. The Framingham Osteoarthritis Study. Arthritis Rheum 1995;38(10):1500−5.
2. Abadie E, Ethgen D, Avouac B, Bouvenot G, Branco J, Bruyere O, *et al*. Recommendations for the use of new methods to assess the efficacy of disease-modifying drugs in the treatment of osteoarthritis. Osteoarthritis Cartilage 2004;12(4):263−8.
3. Altman RD, Abadie E, Avouac B, Bouvenot G, Branco J, Bruyere O, *et al*. Total joint replacement of hip or knee as an outcome measure for structure modifying trials in osteoarthritis. Osteoarthritis Cartilage 2005;13(1):13−9.
4. Eckstein F, Schnier M, Haubner M, Priebsch J, Glaser C, Englmeier KH, *et al*. Accuracy of cartilage volume and thickness measurements with magnetic resonance imaging. Clin Orthop Relat Res 1998; (352):137−48.
5. Ding C, Garnero P, Cicuttini F, Scott F, Cooley H, Jones G. Knee cartilage defects: association with early radiographic osteoarthritis, decreased cartilage volume, increased joint surface area and type II collagen breakdown. Osteoarthritis Cartilage 2005;13(3): 198−205.
6. Burgkart R, Glaser C, Hyhlik-Durr A, Englmeier KH, Reiser M, Eckstein F. Magnetic resonance imaging-based assessment of cartilage loss in severe osteoarthritis: accuracy, precision, and diagnostic value. Arthritis Rheum 2001;44(9):2072−7.
7. Peterfy CG, Guermazi A, Zaim S, Tirman PF, Miaux Y, White D, *et al*. Whole-Organ Magnetic Resonance Imaging Score (WORMS) of the knee in osteoarthritis. Osteoarthritis Cartilage 2004;12(3):177−90.
8. Peterfy C, Li J, Zaim S, Duryea J, Lynch J, Miaux Y, *et al*. Comparison of fixed-flexion positioning with fluoroscopic semi-flexed positioning for quantifying radiographic joint-space width in the knee: test−retest reproducibility. Skeletal Radiol 2003;32(3):128−32.
9. Kothari M, Guermazi A, von Ingersleben G, Miaux Y, Sieffert M, Block JE, *et al*. Fixed-flexion radiography of

the knee provides reproducible joint space width measurements in osteoarthritis. Eur Radiol 2004;14(9):1568–73.
10. Duryea J, Jiang Y, Zakharevich M, Genant HK. Neural network based algorithm to quantify joint space width in joints of the hand for arthritis assessment. Med Phys 2000;27(5):1185–94.
11. Fernandez-Madrid F, Karvonen RL, Teitge RA, Miller PR, Negendank WG. MR features of osteoarthritis of the knee. Magn Reson Imaging 1994;12(5):703–9.
12. Boegard T, Rudling O, Petersson IF, Jonsson K. Correlation between radiographically diagnosed osteophytes and magnetic resonance detected cartilage defects in the patellofemoral joint. Ann Rheum Dis 1998;57(7):395–400.
13. Link TM, Steinbach LS, Ghosh S, Ries M, Lu Y, Lane N, *et al*. Osteoarthritis: MR imaging findings in different stages of disease and correlation with clinical findings. Radiology 2003;226(2):373–81.
14. Cicuttini FM, Wang YY, Forbes A, Wluka AE, Glisson M. Comparison between patella cartilage volume and radiological assessment of the patellofemoral joint. Clin Exp Rheumatol 2003;21(3):321–6.
15. Cicuttini FM, Wluka AE, Forbes A, Wolfe R. Comparison of tibial cartilage volume and radiologic grade of the tibiofemoral joint. Arthritis Rheum 2003;48(3): 682–8.
16. Jones G, Ding C, Scott F, Glisson M, Cicuttini F. Early radiographic osteoarthritis is associated with substantial changes in cartilage volume and tibial bone surface area in both males and females. Osteoarthritis Cartilage 2004;12(2):169–74.
17. Cicuttini FM, Wluka AE, Hankin J, Stuckey S. Comparison of patella cartilage volume and radiography in the assessment of longitudinal joint change at the patellofemoral joint. J Rheumatol 2004;31(7): 1369–72.
18. Raynauld JP, Martel-Pelletier J, Berthiaume MJ, Labonte F, Beaudoin G, de Guise JA, *et al*. Quantitative magnetic resonance imaging evaluation of knee osteoarthritis progression over two years and correlation with clinical symptoms and radiologic changes. Arthritis Rheum 2004;50(2):476–87.
19. Mazzuca SA, Brandt KD, Buckwalter KA, Lequesne M. Pitfalls in the accurate measurement of joint space narrowing in semiflexed, anteroposterior radiographic imaging of the knee. Arthritis Rheum 2004;50(8): 2508–15.
20. Cicuttini FM, Jones G, Forbes A, Wluka AE. Rate of cartilage loss at two years predicts subsequent total knee arthroplasty: a prospective study. Ann Rheum Dis 2004;63(9):1124–7.
21. Cicuttini F, Hankin J, Jones G, Wluka A. Comparison of conventional standing knee radiographs and magnetic resonance imaging in assessing progression of tibiofemoral joint osteoarthritis. Osteoarthritis Cartilage 2005; 13(8):722–7.